

Figure 11: Each individual plot shows the distribution of tracing curve peaks (the argmax layer) across datapoints, using a different tracing window size. Together, the plots show how the distribution of layers where the tracing curves peak for each point shifts outward toward the first and last layer of the model as the tracing window size declines. This is primarily due to a clipping effect from using a window size greater than 1. The way tracing values are computed, a window size of 10 implies that the effect for "layer 1" is from restoring layers 1-5, while the effect for layer "layer 5" is 1-10. As a result, a tracing window size of 10 favors layer 5 over layers 1-4, and reducing the tracing window size leads to these clumps of effects shifting from layer 5 toward layer 1 (and from layer 24 to layer 28)

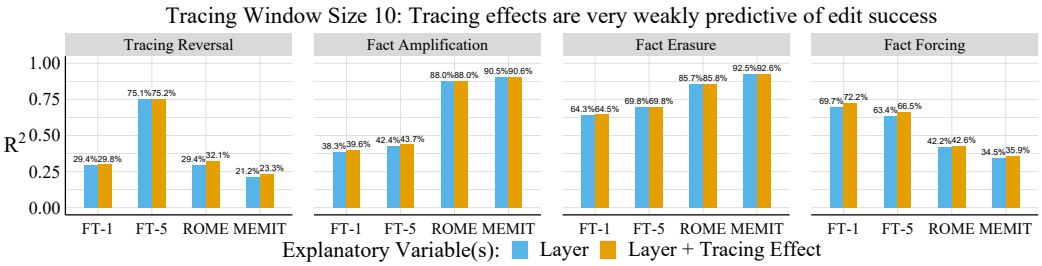


Figure 12: The results of our R^2 analysis for predicting rewrite score are nearly identical between using a tracing window size of 5 (shown in Fig. 6) or 10 (shown here).

values of the tracing window size (Fig. 12), (3) GPT2-XL rather than GPT-J (Fig. 13), (4) the original unscaled metrics from Meng et al. [21] (Fig. 14), and (5) using the tracing effect at the last subject token rather than the max across tokens (Fig. 16). We consider the last subject token effect since this corresponds more directly to the motivation for ROME (see Meng et al. [21]). We expand on each of these experiments below:

Results for Paraphrase, Neighborhood, Overall Metrics. We recreate our regression-based analysis across editing problem variants and editing methods using paraphrase score and neighborhood score as our outcomes rather than Rewrite Score, as well as an Overall Score that is the raw average of the three edit scores. These results are shown in Tables 4, 5, and 6 respectively. Similar to our analysis with rewrite score, these tables show that tracing effects are barely predictive of edit success at all. For paraphrase score, the largest gains in R^2 values are around 0.03 (relative to the layer-only regression model), and for neighborhood score, the largest gain is 0.02. The largest gain for overall score is 0.02 for Fact Forcing with constrained finetuning. Our overall conclusion remains that tracing effects are almost totally unrelated to edit success across editing problem variants, including for different edit success metrics.

Results for Different Tracing Window Sizes. We repeat our analysis from Sec. 5 using tracing effects obtained from a larger tracing window size of 10, to match the value used in Meng et al. [21]. Note that from Fig. 10, we know that the tracing effects grow larger as more adjacent layer representations are restored. When we recreate our main R^2 analysis using tracing effects with window size 10 (shown in Fig. 12), we find that results are nearly identical to those shown in Tables 3, 4, and 5.

Results for GPT2-XL. We rerun our analysis with GPT2-XL, a 48 layer model [30], while editing layers in the range {1, 5, 9, 13, 17, 18, 21, 25, 29, 33, 37, 41, 45, 48}. Here, we use a tracing window size of 10, and we limit our experiments to focus on Error Injection and Fact Forcing editing problems. As seen in Fig. 13, we find very similar trends when explaining rewrite score in terms of the choice

Table 6: R^2 values for predicting **overall** score (raw average of rewrite, paraphrase, and neighborhood scores) from choice of edit layer and tracing effect, across editing problem variants. Diff shows the added effect of including tracing in the regression (Both vs. Layer Only), in terms of R^2, and p-value shows the results from an F-test comparing the Both and Layer Only models. The added effect of including tracing effects is very small across conditions (2% or less).

Ovr. Edit Score		R^2 Values				
Editing Problem	Method	Layer	Trace	Both	Diff	p-value
Error Injection	FT (1 layer)	0.642	0.054	0.643	0.002	0.001
	FT (5 layers)	0.663	0.047	0.665	0.002	0.001
	ROME (1 layer)	0.62	0.003	0.629	0.009	<1e-4
	MEMIT (5 layers)	0.525	0.008	0.534	0.009	<1e-4
Tracing Reversal	FT (1 layer)	0.294	0.025	0.296	0.002	0.054
	FT (5 layers)	0.751	0.045	0.752	0.001	0.032
	ROME (1 layer)	0.296	0.016	0.31	0.014	<1e-4
	MEMIT (5 layers)	0.21	0.036	0.216	0.006	<1e-4
Fact Erasure	FT (1 layer)	0.28	0.007	0.283	0.004	0.008
	FT (5 layers)	0.119	0	0.124	0.004	0.015
	ROME (1 layer)	0.718	0.023	0.718	0	0.729
	MEMIT (5 layers)	0.794	0.025	0.794	0	0.555
Fact Amplification	FT (1 layer)	0.188	0.003	0.199	0.011	<1e-4
	FT (5 layers)	0.224	0.002	0.236	0.013	<1e-4
	ROME (1 layer)	0.583	0.005	0.59	0.007	<1e-4
	MEMIT (5 layers)	0.597	0.005	0.607	0.01	<1e-4
Fact Forcing	FT (1 layer)	0.487	0.056	0.5	0.013	<1e-4
	FT (5 layers)	0.459	0.057	0.475	0.017	<1e-4
	ROME (1 layer)	0.285	0.004	0.291	0.006	<1e-4
	MEMIT (5 layers)	0.226	0.017	0.227	0.001	0.419

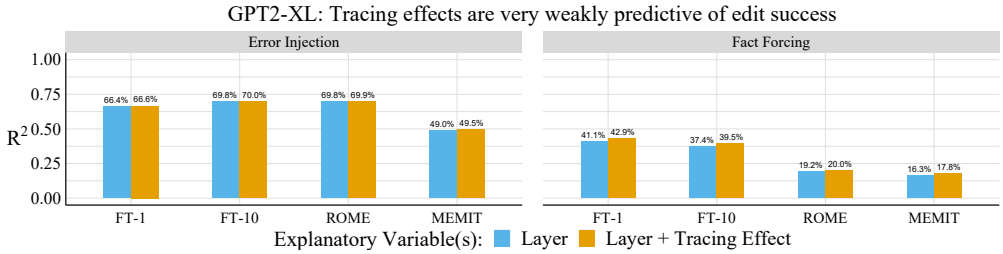


Figure 13: Like with GPT-J, tracing effects are very weakly predictive of edit success across editing problem variants for **GPT2-XL** while Fact Forcing shows the largest relationship. Relative to the R^2 of a model predicting rewrite score based on the choice of edit layer (blue), a model with edit layer and tracing effects (orange) improves the R^2 by at most .02 points for Fact Forcing. The choice of edit layer explains a much greater share of the variance in rewrite score.

of edit layer and the tracing effect at that layer. The largest explanatory effects in terms of R^2 are observed for Fact Forcing with constrained finetuning, but these effects remain small at about 2%.

Results for Unscaled Metrics. We repeat our analysis using the original editing metrics and absolute tracing effects from Meng et al. [21]. Their rewrite magnitude is the absolute difference between the probability of the new target o_{false} and the old true target o_{true} after editing, $p_{\theta^*}(o_{false}|s,r) - p_{\theta^*}(o_{true}|s,r)$. The tracing effect is the absolute tracing effect, $p_\theta(o_{true}|s_{noise}, r, v_{(t,\ell)}) - p_\theta(o_{true}|s_{noise}, r)$, measured at the last subject token index. We adjusted our rewrite and tracing metrics to (1) rely only on the target output probability, rather than difference in probabilities of two different targets which might not be appropriate for our different editing problems, and (2) to always fall between 0 and 1 for better comparability between datapoints, since absolute tracing effect are bounded by the original model probabilities. However, we reach the same conclusions from our analysis when using the original editing metrics. We show an example for rewrite magnitude and the

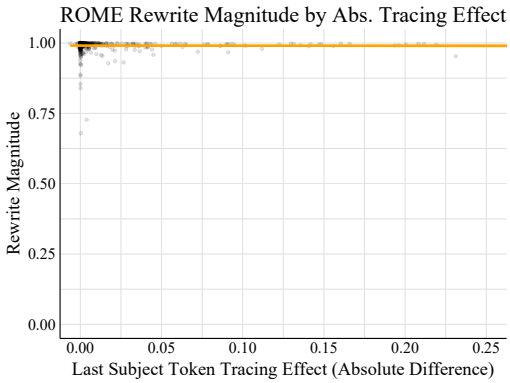


Figure 14: Editing vs. tracing results for ROME at layer 6 for Error Injection, using the un-rescaled rewrite and tracing metrics from Meng et al. [21]. Here, rewrite magnitude is the difference between the probability of the new target o_{false} and the old true target o_{true} after editing, $p_{\theta^*}(o_{false}|s,r) - p_{\theta^*}(o_{true}|s,r)$. The tracing effect is the absolute tracing effect, $p_\theta(o_{true}|s_{noise},r,v_{(t,\ell)}) - p_\theta(o_{true}|s_{noise},r)$, measured at the last subject token index. The correlation here is near zero, at $\rho = -.006$.

Edit Metric	Regression Metric	Predictor(s)	Value
Rewrite Score	R^2	Layer	0.947
		Tracing Effect	0.016
	RMSE	Layer	0.073
		Tracing Effect	0.315
	MAE	Layer	0.02
		Tracing Effect	0.206
Overall Score	R^2	Layer	0.618
		Tracing Effect	0.003
	RMSE	Layer	0.133
		Tracing Effect	0.216
	MAE	Layer	0.11
		Tracing Effect	0.183

Table 7: **Additional regression error metrics** (for CounterFact and ROME) lead us to the same conclusion as our analysis based on R^2. RMSE is root mean squared error, and MAE is mean absolute error. Regressions predicting rewrite score (or overall score) from the choice of edit layer achieve much lower prediction errors than regressions using the tracing effect, suggesting that the choice of edit layer is much more important for edit success than the tracing effect.

absolute tracing effect for Error Injection in Fig. 14. The correlation between edit success and tracing effect is still near zero.

Results for Last Subject Token Effect. ROME increases the target probability $p(o_{false}|s,r)$ by optimizing for a new output representation from a chosen MLP layer *at the last subject token index*. Meng et al. [21] show that this choice of token representation is critical to the success of the editing method, which is a hypothesis directly motivated by the results from their Causal Tracing analysis. In our paper, we by default report results using tracing effects that are the max across tokens at a given layer, for better comparability across the editing methods we use. However, when we repeat our analysis using the tracing effect specifically at the last subject token index, we obtain the same negative conclusions about the relationship between Causal Tracing localization and ROME editing performance. We show the correlations between Rewrite Score and Last Subject Token Tracing Effect in Fig. 16, where we see there are no positive correlations between editing success and tracing results at any layer in GPT-J.